

App Dev-II Project Report

1. Student Details

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About Me: As a Data Science student, I focus on creating strong backend systems that don't crash when lots of data is processed. I like working on tools that automatically run tasks in the background, ensuring a clean and lag-free experience for users.

2. Project Details

Project Title: Placement Portal Cell

Problem Statement:

To build a dynamic Placement Portal that helps colleges, companies, and students manage campus hiring easily. The goal is to automate heavy operations like exporting applicant data and sending automated notifications in the background, keeping the website fast and crash-free for all users.

Approach:

The portal is built using a modular structure with Flask for backend routing and a responsive frontend layout. To prevent the server from freezing during heavy tasks like exporting student applicant data, the system offloads operations to Celery background workers using a Redis broker. This approach ensures that data processing and automated reminders run smoothly in the background without affecting the user experience.

3. AI/LLM Declaration

I used ChatGPT to help understand complex backend architecture concepts, improve the front-end UI layouts, and assist slightly with code implementation and understanding the Error.

The total AI usage was limited to about 10-15% for conceptual guidance and styling suggestions, while the core project logic, debugging, and system integration were completed manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework.
Flask Blueprints	To create a modular, clean, and organized application structure.
Vue 2	Frontend framework for building reactive and dynamic user layouts.
Celery	For running asynchronous background tasks like automated reports.
Redis	High-speed message broker for Celery and application caching engine.
SQLAlchemy / SQLite	Database management and Object Relational Mapping for data models.
Flask-Mail / Mail Engine	To handle and trigger automated interview reminders and emails.

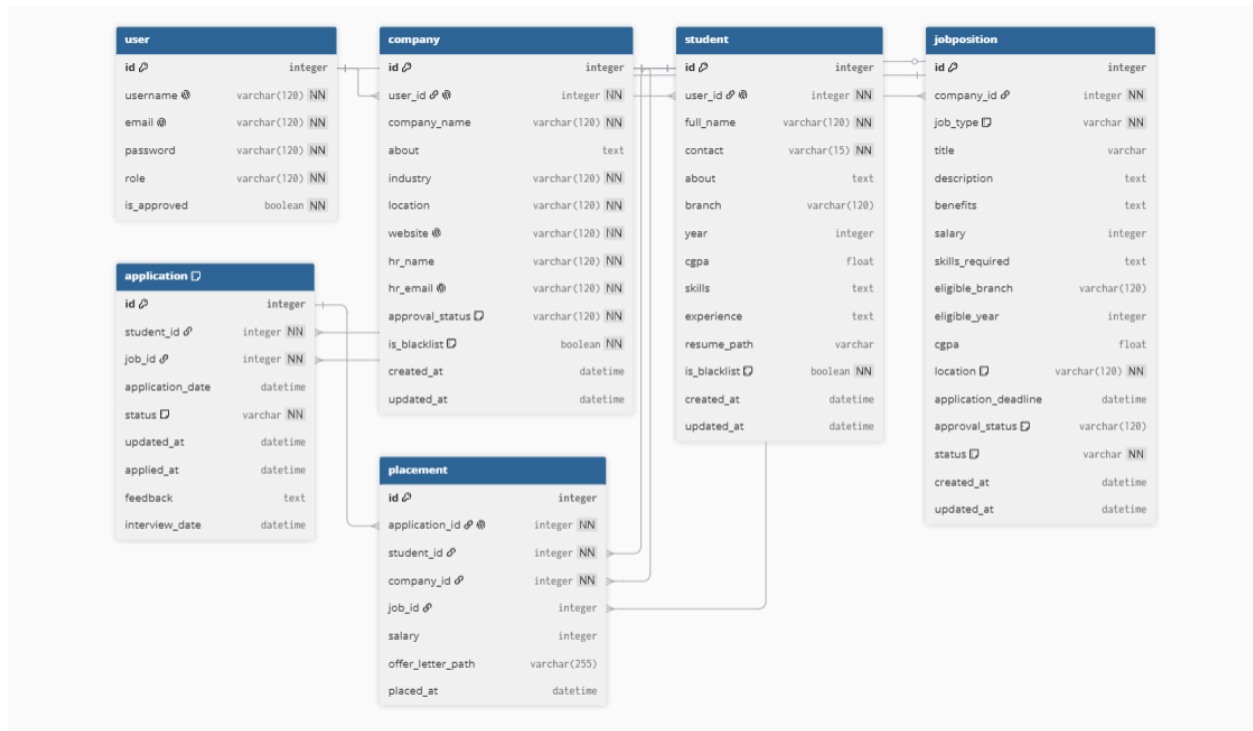
5. Database Schema / ER Diagram

Tables:

- **User** — stores user account details such as username, email, password, role, and approval status.
- **Company** — stores company profile details such as company name, industry, location, website, HR details, and approval status.
- **Student** — stores student profile details such as name, contact, branch, year, CGPA, skills, experience, and resume.
- **JobPosition** — stores job details such as company, job type, title, salary, required skills, eligibility criteria, location, and application deadline.
- **Application** — stores student job applications, including application date, status, feedback, and interview date.
- **Placement** — stores final placement details, including student, company, job, salary, offer letter, and placement date.

Relationships:

- One-to-One → User → Company
- One-to-One → User → Student
- One-to-Many → Company → JobPosition
- One-to-Many → Student → Application
- One-to-Many → JobPosition → Application
- One-to-One → Application → Placement
- One-to-Many → Student → Placement
- One-to-Many → Company → Placement
- One-to-Many → JobPosition → Placement



7. Video Presentation :

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